

Neck pain — cervical spine

NOVA-EXO+ 3 cc + NOVA-FLEX 2 cc · physician treatment protocol

A standardized framework for chronic neck pain of cervical origin. Exosome volume is placed bilaterally into the upper trapezius, where paraspinal muscle guarding and myofascial pain are generated, and a single intramuscular NOVA-FLEX dose is given in the deltoid of the non-dominant extremity.

Indications

- Chronic neck pain > 3 months of cervical origin
- Cervical facet arthropathy or degenerative disc disease
- Cervicogenic headache with upper trapezius involvement
- Myofascial pain and trigger points in the upper trapezius
- Postural or occupational strain unresponsive to conservative care
- Failed PT, activity modification, medication, or prior injections

Patient preparation

1. Verify indication; review cervical imaging when available.
2. Document a baseline neurologic exam of both upper extremities.
3. Confirm and mark the dominant extremity; obtain informed consent.
4. Record baseline VAS, Neck Disability Index, and cervical ROM.
5. Position seated with the neck in slight flexion, shoulders relaxed.
6. Prepare skin with chlorhexidine; establish sterile field.

POST-PROCEDURE CARE

- Observe 15 minutes following injection
- Soreness at the trapezius and deltoid sites for 24–72 h
- Avoid NSAIDs and alcohol for 72 h when appropriate
- No heavy lifting or overhead work for 48 h; gentle cervical ROM encouraged

Contraindications

- Active local or systemic infection
- Pregnancy or breastfeeding
- Progressive myelopathy or radiculopathy with motor deficit
- Cervical instability, fracture, or recent instrumentation
- Corticosteroid injection within 4 weeks
- Uncontrolled bleeding disorder or significant anticoagulation risk

Injection & distribution

TOTAL VOLUME · 3 INJECTIONS		5 cc
1	NOVA-EXO+ Right upper trapezius, lateral to the cervical spine	1.5 cc
2	NOVA-EXO+ Left upper trapezius, lateral to the cervical spine	1.5 cc
3	NOVA-FLEX (MSC) Deltoid IM, non-dominant extremity	2 cc

Angle trapezius needles away from midline into the muscle belly. Aspirate at every site. 25 G trapezius, 23 G deltoid.

FOLLOW-UP & OUTCOME MEASURES

Week 2

Safety review

Week 6

Functional progress

Week 12

Primary outcome

VAS PAIN

NECK DISABILITY INDEX

CERVICAL ROM

HEADACHE DAYS

SLEEP

ANALGESIC USE

ADVERSE EVENTS

REGULATORY

Intended for physician-directed clinical and investigational use. Providers are responsible for compliance with applicable federal, state, facility, IRB, and professional practice requirements.

Today you received injections of an exosome product into the muscles at the top of each shoulder, plus one intramuscular cell injection in your upper arm. The products used are made from donated birth tissue and are not FDA-approved treatments. This page tells you what is normal in the days ahead, how to care for yourself, and when to call us.

DATE OF TREATMENT

SITE(S) TREATED

NEXT VISIT

WHAT TO EXPECT

Changes are gradual — do not judge the result in the first days

FIRST 72 HOURS

Soreness at the trapezius and arm sites is common for one to three days. Your usual neck ache may briefly flare before it eases.

WEEKS 1–2

Keep the neck gently moving. We will call or see you around week 2 to make sure everything settled as expected.

WEEKS 6–12

Progress is measured at weeks 6 and 12 against your baseline pain, function, and range-of-motion scores.

DO

- Keep gentle neck range of motion going today
- Use cold packs on sore spots 10–15 minutes at a time
- Take acetaminophen (Tylenol) for soreness if needed
- Keep your week 2, 6, and 12 visits

AVOID

- Anti-inflammatory pain relievers for 72 hours
- Alcohol for 72 hours
- Heavy lifting and overhead work for 48 hours
- Deep-tissue massage of the injected areas for 1 week

CALL US RIGHT AWAY IF

For an emergency, call 911 first

- New weakness, numbness, or tingling in an arm or hand
- A severe or unusual headache, or a stiff neck with fever
- Fever above 101 °F or spreading redness at a site
- Trouble swallowing or breathing

CLINIC PHONE

AFTER-HOURS PHONE

NOTES FROM YOUR VISIT